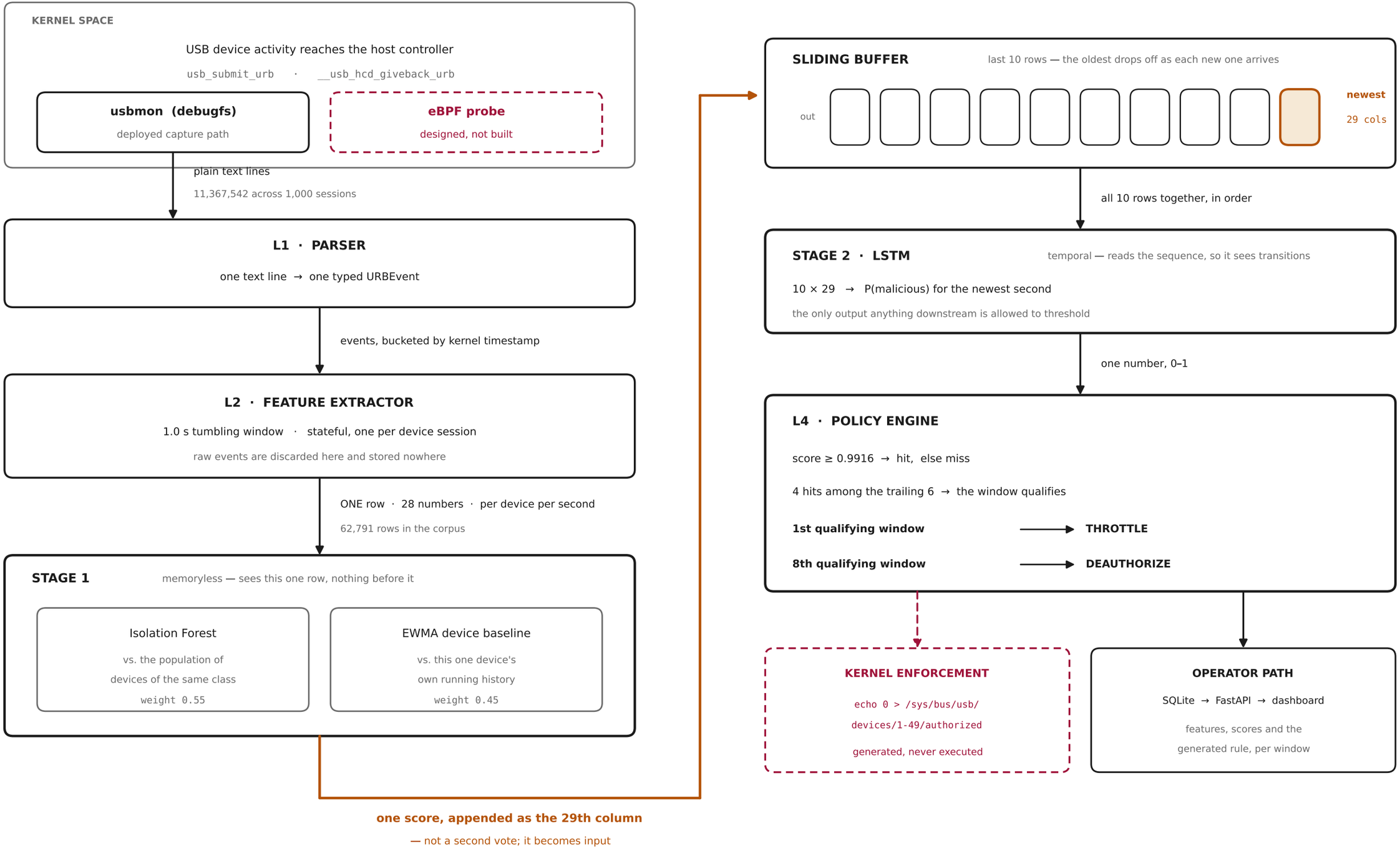
KERNEL SPACE
USB device activity reaches the host controller
usb_submit_urb · __usb_hcd_giveback_urb
usbmon (debugfs)
deployed capture path
eBPF probe
designed, not built
plain text lines
11,367,542 across 1,000 sessions
L1 · PARSER
one text line → one typed URBEvent
events, bucketed by kernel timestamp
L2 · FEATURE EXTRACTOR
1.0 s tumbling window · stateful, one per device session
raw events are discarded here and stored nowhere
ONE row · 28 numbers · per device per second
62,791 rows in the corpus
STAGE 1
memoryless — sees this one row, nothing before it
Isolation Forest
vs. the population of
devices of the same class
weight 0.55
EWMA device baseline
vs. this one device's
own running history
weight 0.45
one score, appended as the 29th column
— not a second vote; it becomes input
SLIDING BUFFER
last 10 rows — the oldest drops off as each new one arrives
out
newest
29 cols
all 10 rows together, in order
STAGE 2 · LSTM
temporal — reads the sequence, so it sees transitions
10 × 29 → P(malicious) for the newest second
the only output anything downstream is allowed to threshold
one number, 0–1
L4 · POLICY ENGINE
score ≥ 0.9916 → hit, else miss
4 hits among the trailing 6 → the window qualifies
1st qualifying window → THROTTLE
8th qualifying window → DEAUTHORIZE
KERNEL ENFORCEMENT
echo 0 > /sys/bus/usb/
devices/1-49/authorized
generated, never executed
OPERATOR PATH
SQLite → FastAPI → dashboard
features, scores and the
generated rule, per window
data flow
the relay — Stage 1's output is Stage 2's input
designed but not built / generated but not executed